

Frequency characteristics

Bode plots

Circuits Theory

Transfer function in s-domain: $H(s) = \frac{U_{OUT}(s)}{U_{IN}(s)}$

Substitution: $s \Rightarrow j\omega$

Complex Transfer function: $H(s) \Rightarrow H(\omega) \in \mathbb{C}$

1. Magnitude characteristic: $A = 20 \log_{10} |H(\omega)|$

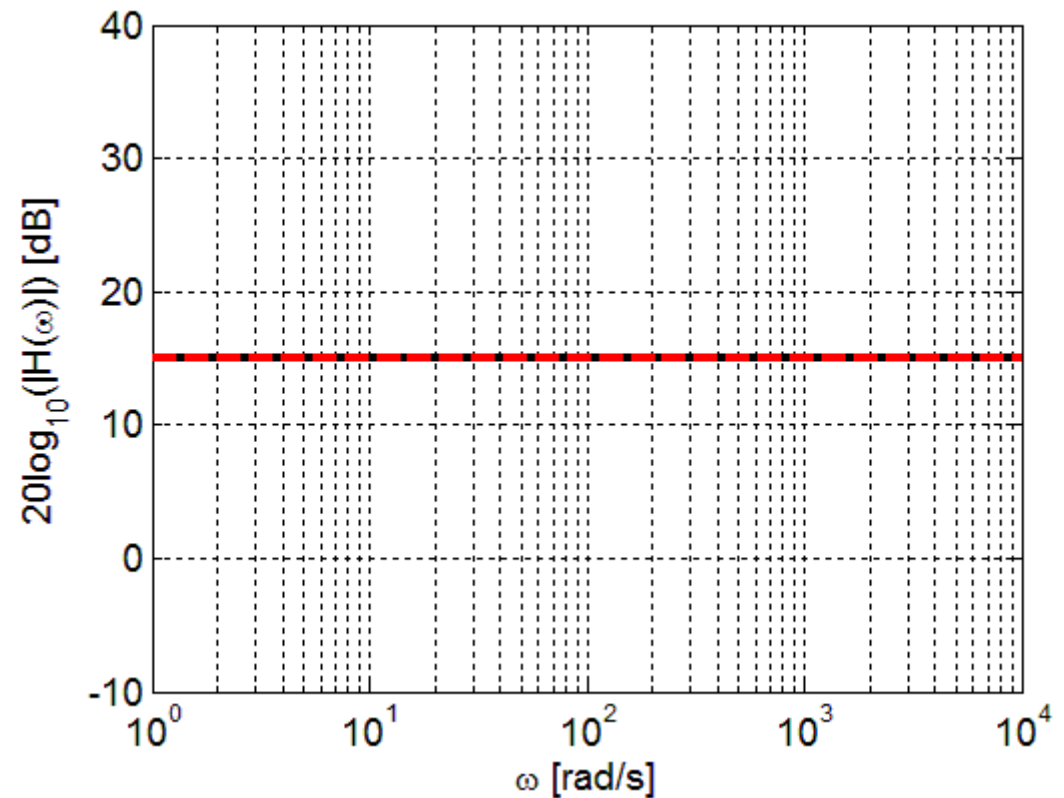
2. Phase characteristic: $\Phi = \arg[H(\omega)]$

Approximations can be easily plotted using logarithmic scale

Magnitude characteristics

Transfer function in s-domain: $H(s) = k = \text{const}$

$$A(\omega) = 20\log_{10}(k)$$



Transfer function in s-domain: $H(s) = \left(\frac{s}{s_0} + 1 \right)$

$$H(\omega) = \left(\frac{j\omega}{s_0} + 1 \right)$$

$$|H(\omega)| = \sqrt{\left(\frac{\omega}{s_0} \right)^2 + 1}$$

$$A(\omega) = 20 \log_{10} |H(\omega)| = 20 \log_{10} \sqrt{\left(\frac{\omega}{s_0} \right)^2 + 1}$$

$$A(\omega) = 10 \log_{10} \left[\left(\frac{\omega}{s_0} \right)^2 + 1 \right]$$

Bode approximation of $A(\omega) = 10 \log_{10} \left[\left(\frac{\omega}{s_0} \right)^2 + 1 \right]$

$$\omega < s_0 \Rightarrow \left(\frac{\omega}{s_0} \right)^2 + 1 \approx 1$$

$$A(\omega) = 10 \log_{10}(1) = 0$$

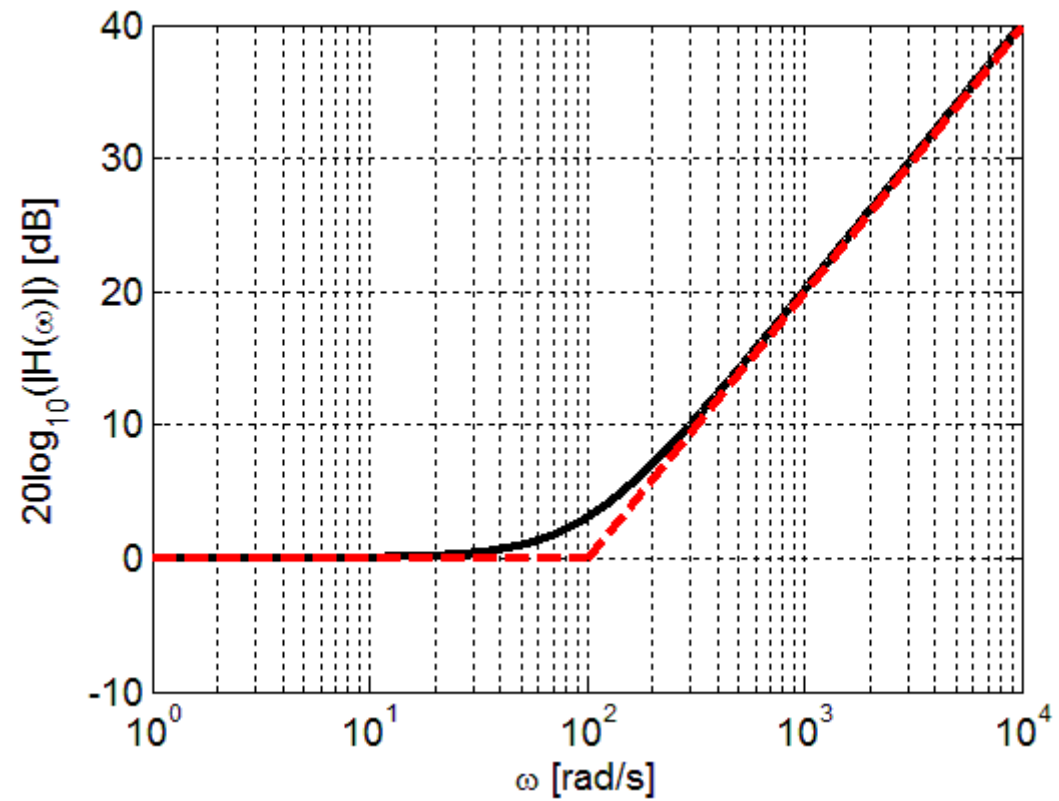
$$\omega > s_0 \Rightarrow \left(\frac{\omega}{s_0} \right)^2 + 1 \approx \left(\frac{\omega}{s_0} \right)^2$$

$$A(\omega) = 10 \log_{10} \left(\frac{\omega}{s_0} \right)^2 = 20 \log_{10} \left(\frac{\omega}{s_0} \right) = 20 \log_{10}(\omega) - 20 \log_{10}(s_0)$$

$$A(\omega) = 10 \log_{10} \left[\left(\frac{\omega}{s_0} \right)^2 + 1 \right]$$

3-dB error for $\omega = s_0$

Slope: 20 dB/decade

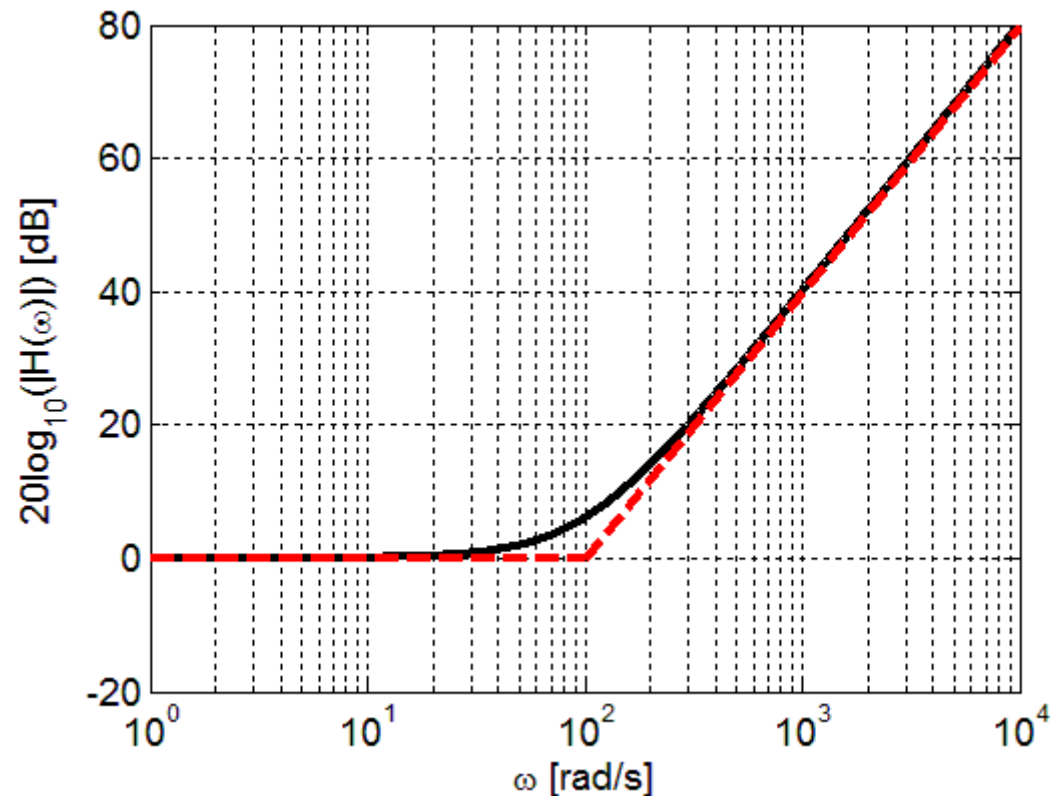


Transfer function in s-domain: $H(s) = \left(\frac{s}{s_0} + 1 \right)^n$

$$A(\omega) = 10n \log_{10} \left[\left(\frac{\omega}{s_0} \right)^2 + 1 \right]$$

$3n$ -dB error for $\omega = s_0$

Slope: $20n$ dB/decade



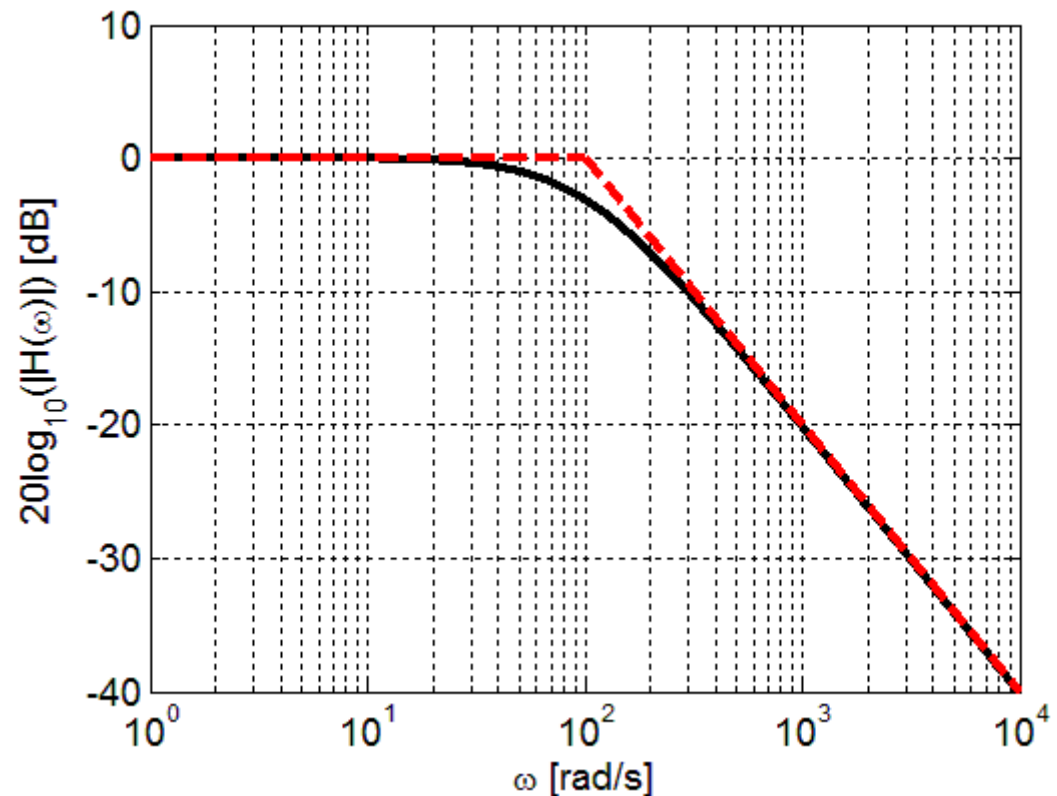
Transfer function in s-domain: $H(s) = \frac{1}{\frac{s}{s_0} + 1}$

$$H(s) = \frac{1}{\frac{s}{s_0} + 1} = \left(\frac{s}{s_0} + 1 \right)^{-1}$$

$$A(\omega) = -10 \log_{10} \left[\left(\frac{\omega}{s_0} \right)^2 + 1 \right]$$

3-dB error for $\omega = s_0$

Slope: -20 dB/decade



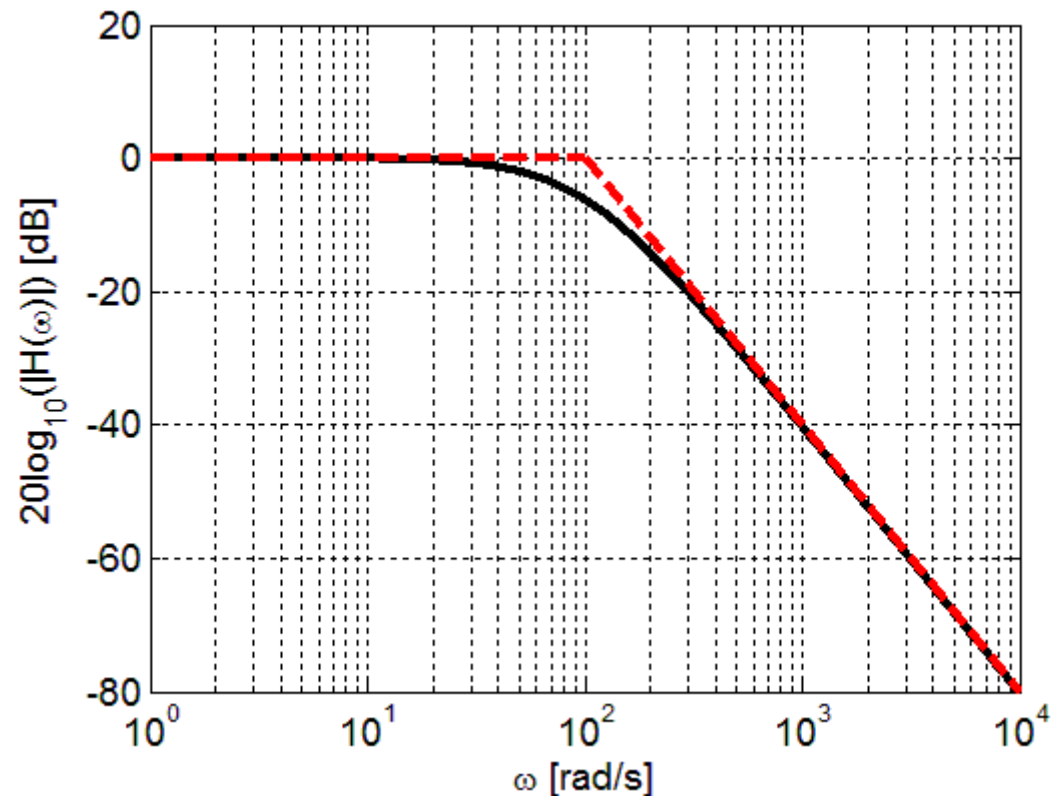
Transfer function in s-domain: $H(s) = \frac{1}{\left(\frac{s}{s_0} + 1\right)^n}$

$$H(s) = \frac{1}{\left(\frac{s}{s_0} + 1\right)^n} = \left(\frac{s}{s_0} + 1\right)^{-n}$$

$$A(\omega) = -10n \log_{10} \left[\left(\frac{\omega}{s_0}\right)^2 + 1 \right]$$

3n-dB error for $\omega = s_0$

Slope: -20n dB/decade

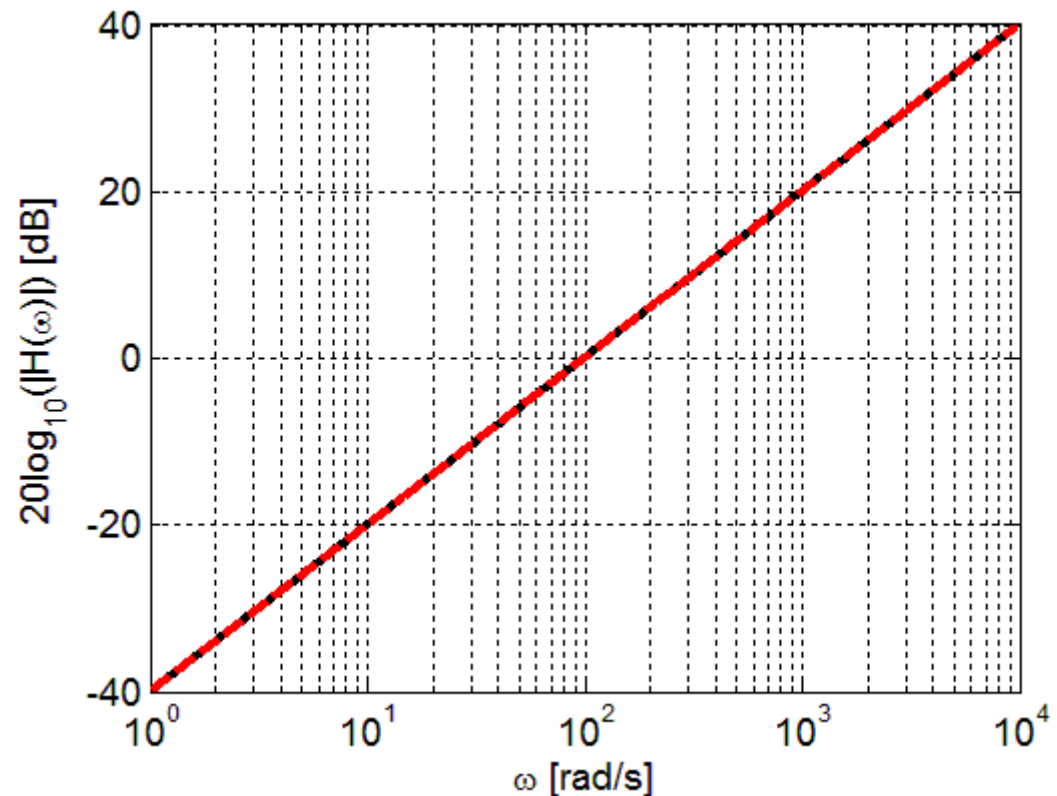


Transfer function in s-domain: $H(s) = \left(\frac{s}{k}\right)^n$

$$A(\omega) = 20n \log_{10}\left(\frac{\omega}{k}\right) = 20n \log_{10}(\omega) - 20n \log_{10}(k)$$

$$A(k) = 0$$

Slope: $20n$ dB/decade

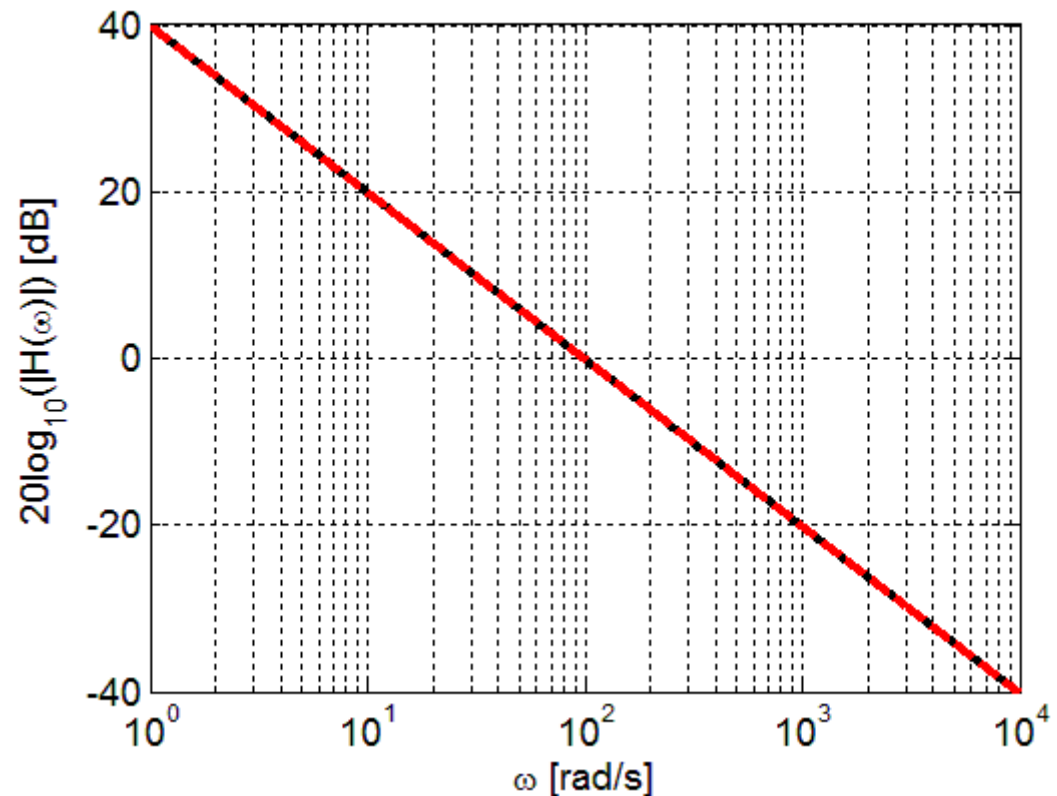


Transfer function in s-domain: $H(s) = \left(\frac{k}{s}\right)^n$

$$A(\omega) = 20n \log_{10} \left(\frac{k}{\omega} \right) = -20n \log_{10}(\omega) + 20n \log_{10}(k)$$

$$A(k) = 0$$

Slope: $-20n$ dB/decade



Example

$$H(s) = \frac{1}{2} \frac{s^2 + 1010s + 10000}{s^2 + 100s}$$

Transfer function in s-domain: $H(s) = \frac{1}{2} \frac{s^2 + 1010s + 10000}{s^2 + 100s}$

$$H(s) = \frac{1}{2} \frac{(s+1000)(s+10)}{s(s+100)} = \frac{10000 \left(\frac{s}{1000} + 1 \right) \left(\frac{s}{10} + 1 \right)}{200s \left(\frac{s}{100} + 1 \right)}$$

$$H(s) = \frac{50}{s} \frac{\left(\frac{s}{1000} + 1 \right) \left(\frac{s}{10} + 1 \right)}{\left(\frac{s}{100} + 1 \right)} = H_1(s)H_2(s)H_3(s)H_4(s)$$

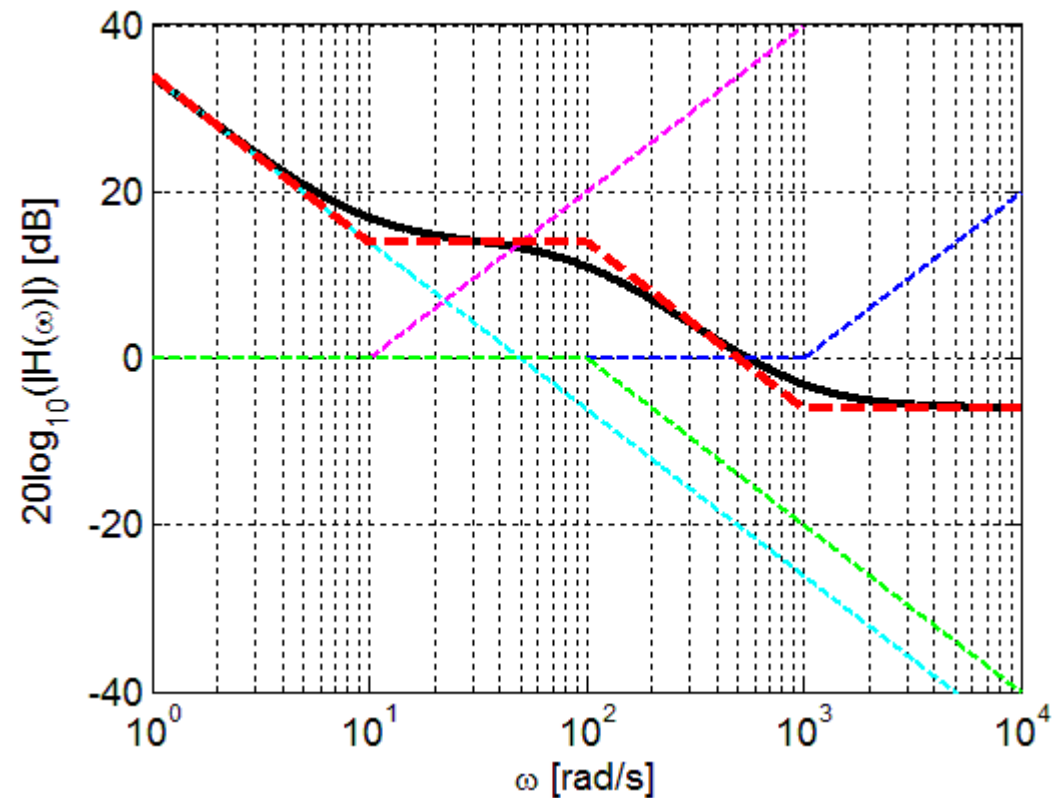
$$H(s) = \frac{50}{s} \frac{\left(\frac{s}{1000} + 1\right) \left(\frac{s}{10} + 1\right)}{\left(\frac{s}{100} + 1\right)} = H_1(s)H_2(s)H_3(s)H_4(s)$$

$$H_1(s) = \frac{50}{s}$$

$$H_2(s) = \frac{s}{1000} + 1$$

$$H_3(s) = \frac{s}{10} + 1$$

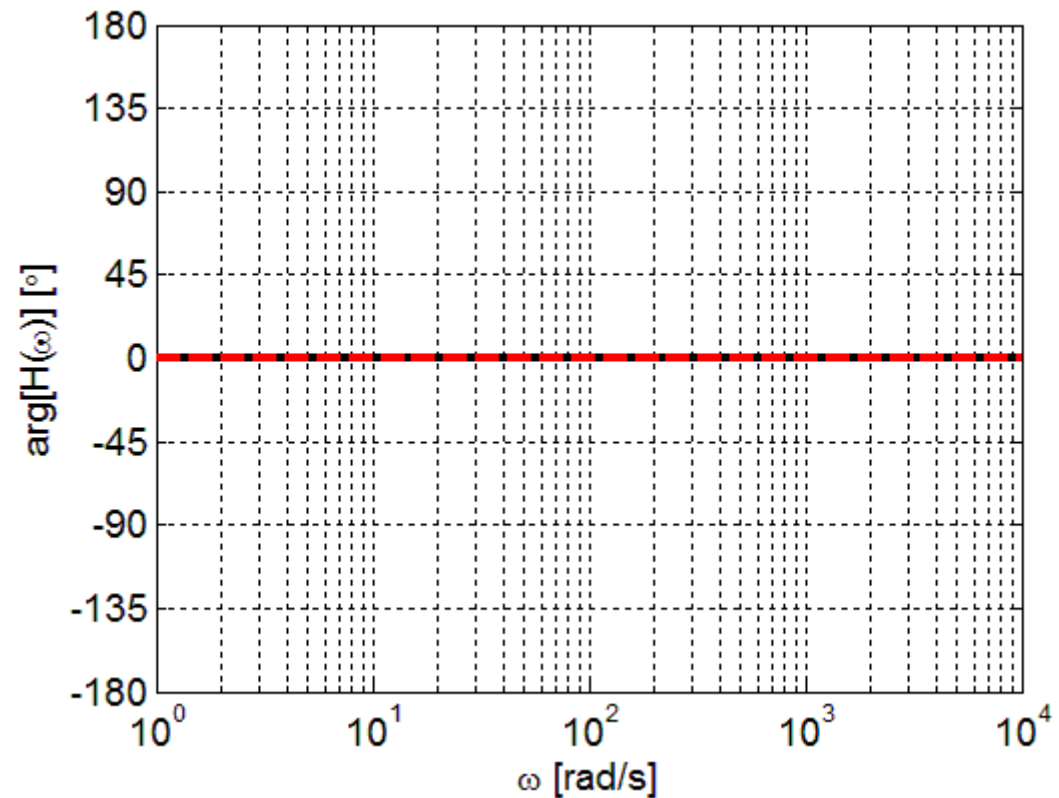
$$H_4(s) = \frac{1}{\left(\frac{s}{100} + 1\right)}$$



Phase characteristics

Transfer function in s-domain: $H(s) = k = \text{const}$

$$\Phi(\omega) = \arg[H(\omega)] = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan\left(\frac{0}{k}\right) = 0^\circ$$



Transfer function in s-domain: $H(s) = \left(\frac{s}{s_0} + 1 \right)$

$$H(\omega) = \left(\frac{j\omega}{s_0} + 1 \right)$$

$$\Phi(\omega) = \arg[H(\omega)] = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan \left(\frac{\omega}{s_0} \right)$$

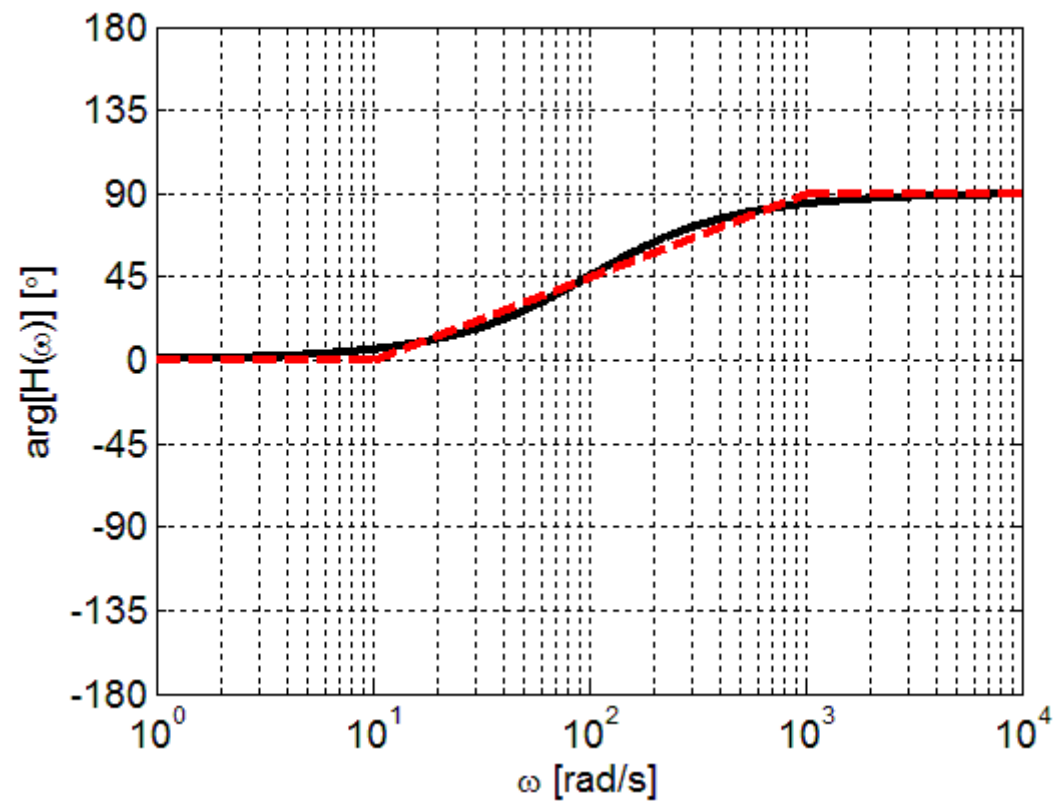
$$\Phi(\omega) = \arg[H(\omega)] = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan\left(\frac{\omega}{s_0}\right)$$

$$\omega \ll s_0 \Rightarrow \frac{\omega}{s_0} \approx 0 \quad \Phi(\omega) = 0$$

$$\omega = s_0 \Rightarrow \frac{\omega}{s_0} = 1 \quad \Phi(\omega) = \arctan(1) = 45^\circ \quad \text{Slope } 45^\circ/\text{decade}$$

$$\omega \gg s_0 \Rightarrow \frac{\omega}{s_0} \rightarrow \infty \quad \Phi(\omega) = \lim_{\omega \rightarrow \infty} \left[\arctan\left(\frac{\omega}{s_0}\right) \right] = 90^\circ$$

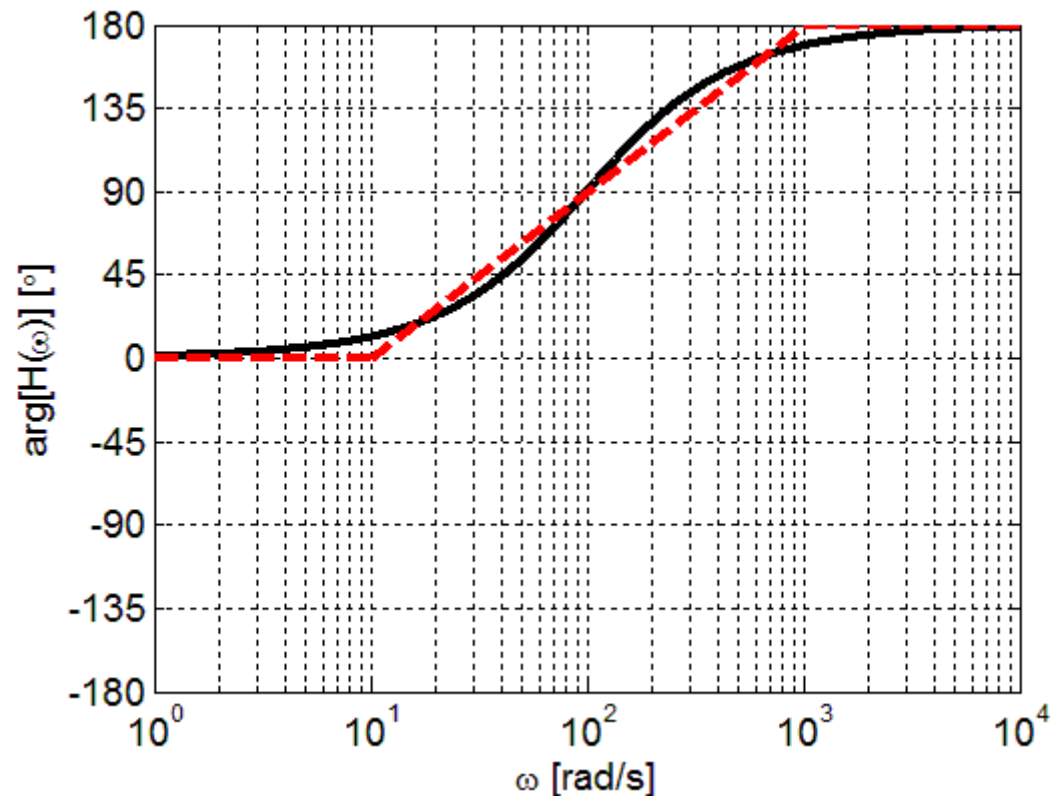
$$\Phi(\omega) = \arg[H(\omega)] = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan\left(\frac{\omega}{s_0}\right)$$



Transfer function in s-domain: $H(s) = \left(\frac{s}{s_0} + 1 \right)^n$

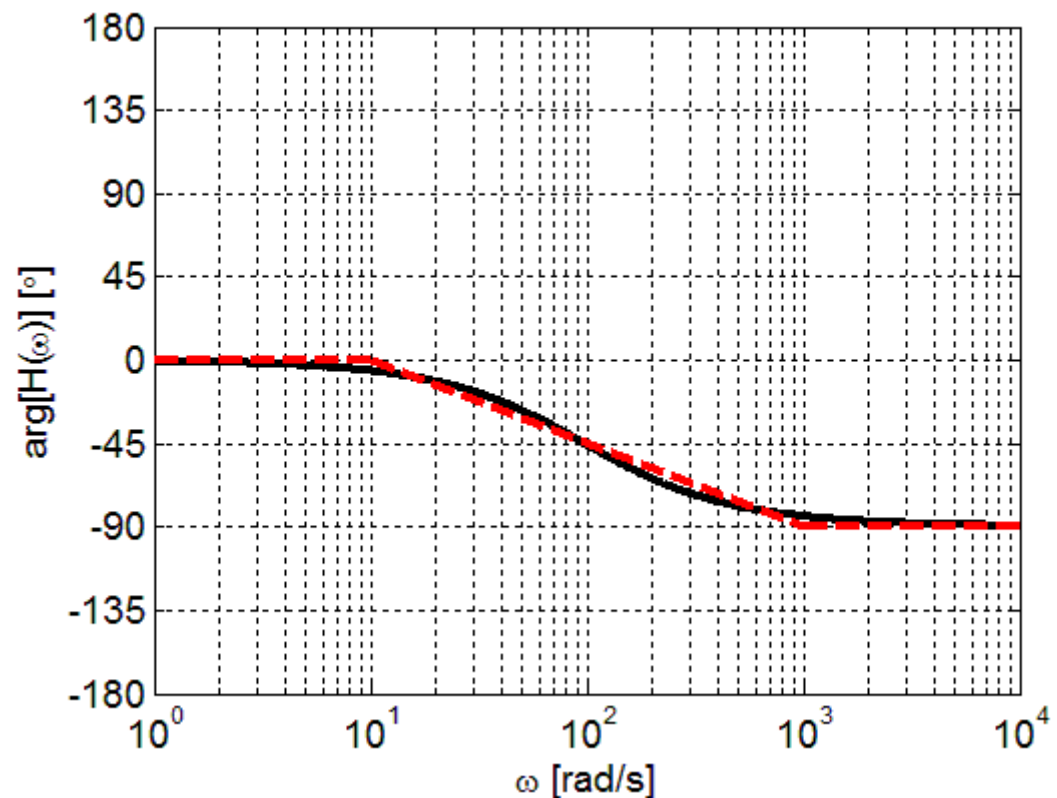
$$\Phi(\omega) = \arg[H(\omega)] = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = n \arctan \left(\frac{\omega}{s_0} \right)$$

For $\omega = s_0$
Slope $n45^\circ/\text{decade}$



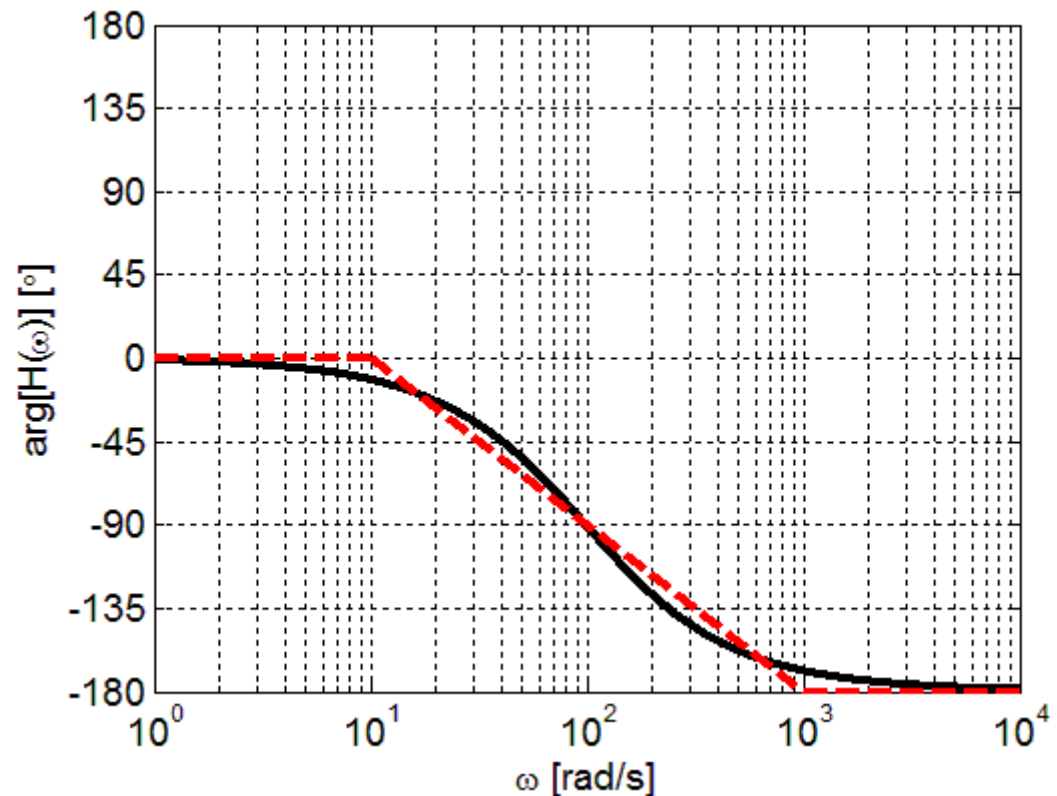
Transfer function in s-domain: $H(s) = \frac{1}{\left(\frac{s}{s_0} + 1\right)}$

$$\Phi(\omega) = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan\left(\frac{0}{1}\right) - \arctan\left(\frac{\omega}{s_0}\right) = -\arctan\left(\frac{\omega}{s_0}\right)$$



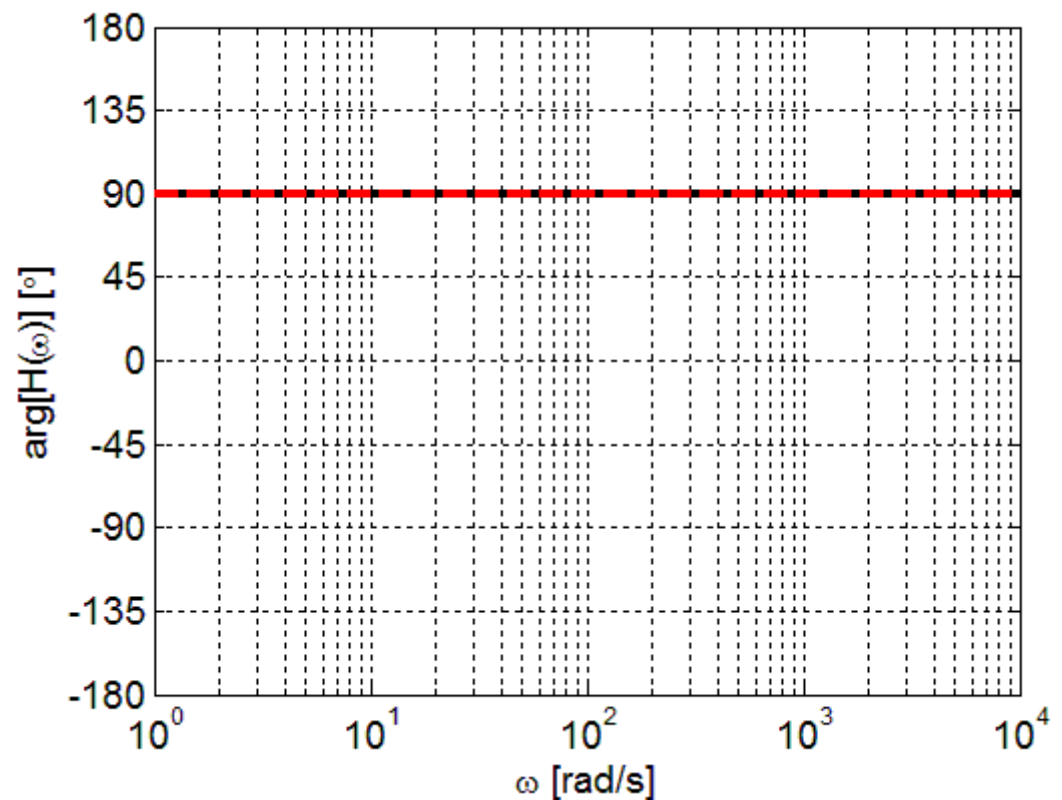
Transfer function in s-domain: $H(s) = \frac{1}{\left(\frac{s}{s_0} + 1\right)^n}$

$$\Phi(\omega) = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \arctan\left(\frac{0}{1}\right) - n \arctan\left(\frac{\omega}{s_0}\right) = -n \arctan\left(\frac{\omega}{s_0}\right)$$



Transfer function in s-domain: $H(s) = \left(\frac{s}{k}\right)^n$

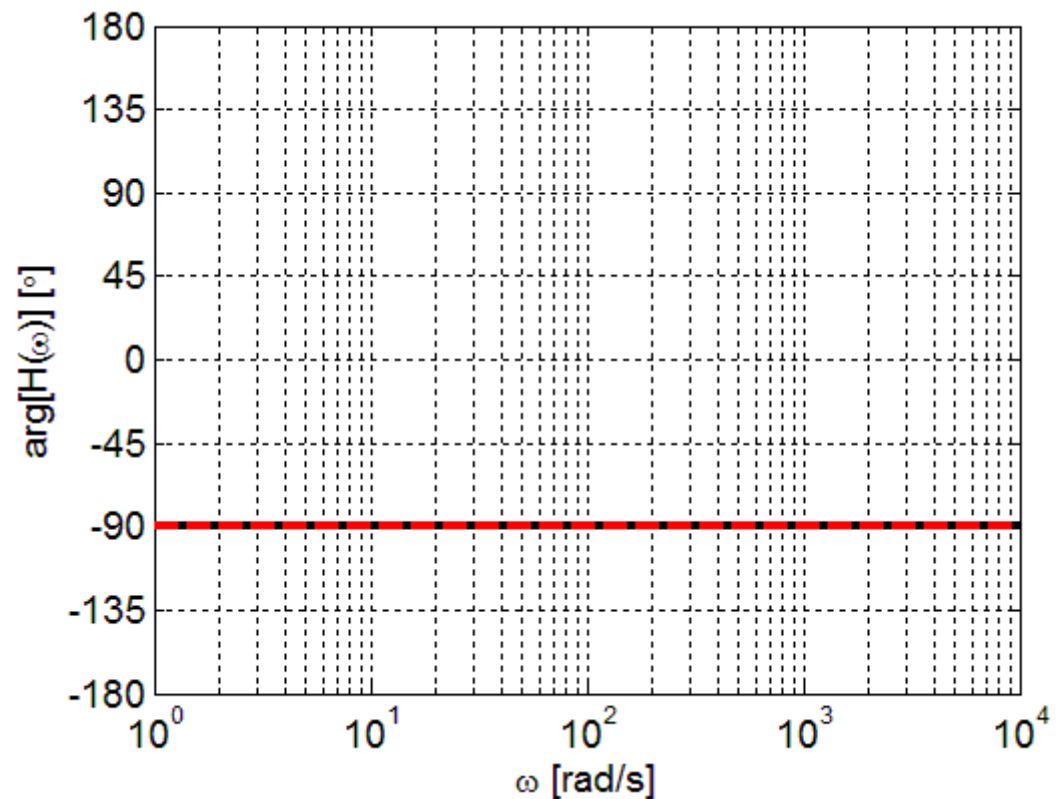
$$\Phi(\omega) = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \lim_{\alpha \rightarrow 0} \arctan \left(\frac{\frac{\omega}{k}}{\alpha} \right) = n90^\circ$$



Transfer function in s-domain: $H(s) = \left(\frac{k}{s}\right)^n$

$$\Phi(\omega) = \arctan \frac{\Im[H(\omega)]}{\Re[H(\omega)]} = \lim_{\alpha \rightarrow 0} \arctan \left(\frac{-\frac{k}{\omega}}{\alpha} \right) = -n90^\circ$$

$$\left(\frac{k}{s} \Rightarrow \frac{k}{j\omega} = -j \frac{k}{\omega} \right)$$



Example

$$H(s) = \frac{1}{2} \frac{s^2 + 1010s + 10000}{s^2 + 100s}$$

Transfer function in s-domain: $H(s) = \frac{1}{2} \frac{s^2 + 1010s + 10000}{s^2 + 100s}$

$$H(s) = \frac{1}{2} \frac{(s+1000)(s+10)}{s(s+100)} = \frac{10000 \left(\frac{s}{1000} + 1 \right) \left(\frac{s}{10} + 1 \right)}{200s \left(\frac{s}{100} + 1 \right)}$$

$$H(s) = \frac{50}{s} \frac{\left(\frac{s}{1000} + 1 \right) \left(\frac{s}{10} + 1 \right)}{\left(\frac{s}{100} + 1 \right)} = H_1(s)H_2(s)H_3(s)H_4(s)$$

$$H(s) = \frac{50}{s} \frac{\left(\frac{s}{1000} + 1\right) \left(\frac{s}{10} + 1\right)}{\left(\frac{s}{100} + 1\right)} = H_1(s)H_2(s)H_3(s)H_4(s)$$

$$H_1(s) = \frac{50}{s}$$

$$H_2(s) = \frac{s}{1000} + 1$$

$$H_3(s) = \frac{s}{10} + 1$$

$$H_4(s) = \frac{1}{\left(\frac{s}{100} + 1\right)}$$

